PREP 3, LESSONS 5–6

# Internal energy and latent heat

Heating graphs; changes of state at constant temperature; $E = mL$

SCORE / 10

Name _____ Class _____ Due _____

## Do Now REMEMBER

From earlier lessons, 1 mark each

1 Define density. [1]

2 Name the method used to measure the volume of an irregularly shaped object. [1]

3 A gas is heated at constant volume. State what happens to its pressure. [1]

## Key ideas LESSONS 5–6

Show your working

4 A solid is heated at a steady rate until it becomes a gas.

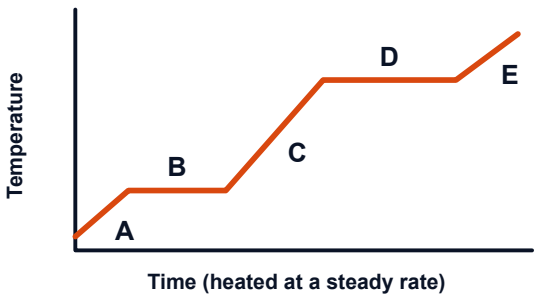
a The section that shows the substance melting is [1]

| A | B | C | D | E |
|---|---|---|---|---|

b In section C the substance is a [1]

| solid | liquid | gas |
|---|---|---|

c Explain why the temperature does not change while the substance melts. [2]




5 Specific latent heat of vaporisation of water = 2 260 000 J/kg. Use $E = mL$

Calculate the energy needed to boil away 0.50 kg of water at 100 °C. [3]

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Hand in on the due date